Examiner only

Answer **all** *questions.*

1. *(a)* In the production of steel alloy, atoms of carbon are added to iron. The resulting alloy is less *ductile* than pure iron. State the meaning of the term *ductile*, and describe, on an atomic scale, why the addition of carbon atoms can make steel less ductile than iron. [3]

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(b) A wire of length 2.4 m and diameter 0.60 mm is made of steel of Young modulus $200 \times 10^9\,\mathrm{N\,m^{-2}}$. The wire is loaded so that its length is increased by 1.8 mm. Assuming that the change is elastic, calculate:

(i) the strain; [1]

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(ii) the applied stress; [2]

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(iii) the force applied to the wire; [2]

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(iv) the elastic energy stored in the wire. [2]

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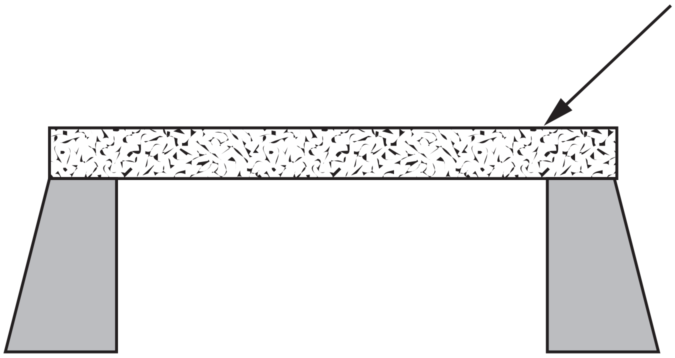
(c) The diagram shows how a gap can be bridged using a concrete beam.



(i) **On the diagram** label a point in tension with the letter **T**, and a point in compression with the letter **C**. [1]

(ii) Inserting a pre-stressed steel bar into the concrete beam would increase the breaking stress of the concrete. **On the diagram**, draw a pre-stressed steel bar in an appropriate position. [1]

(iii) Explain how the steel bar strengthens the beam. [2]

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